

USER MANUAL

ESP32 MiniPLC

Part No. 0000001

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Power connection setup

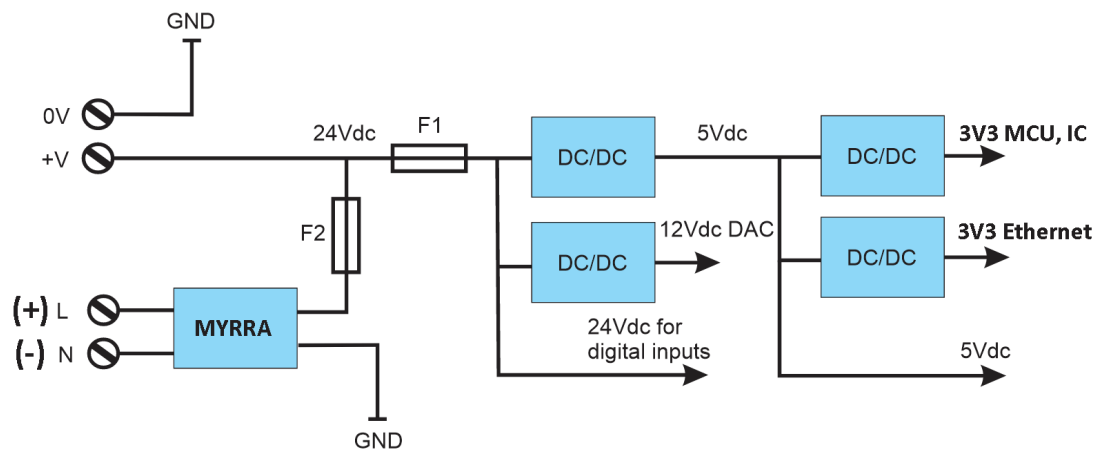
The MiniPLC can be powered either by an external 24VDC power supply or through its internal power supply. It is equipped with an integrated MYRRA 47156 power supply, which accepts 24 V DC at +V/0V, or 85–265 V AC, 47–63 Hz at L/N, operating at a frequency of 47 to 63Hz.

MiniPLC power consumption: typ. 1.85W, max. 3W.

MYRRA 47156 specifications:

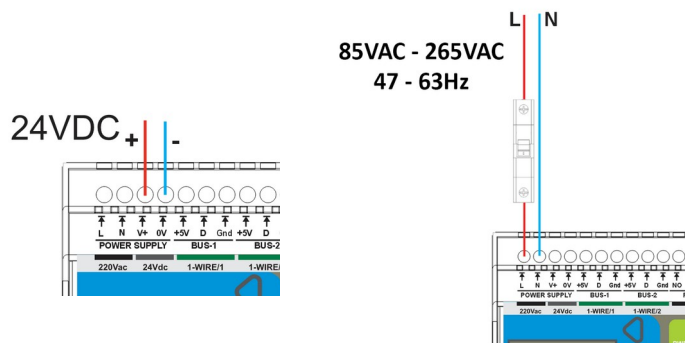
Output voltage (DC Volts)	Output current (DC mA)	Output Power (W)	Efficiency (%)	Ta (°C)
24	220	5	79	+50

The following is a simplified schematic of the internal power distribution:



24 V DC external supply

85–265 V AC via internal power supply

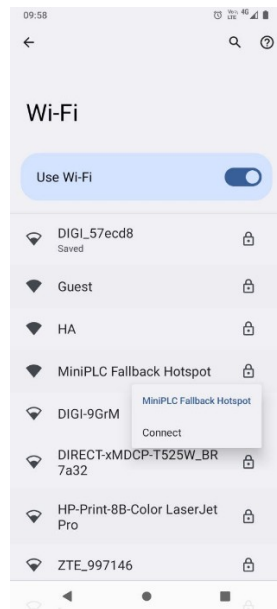


***Do not operate the internal power supply while a 24Vdc external power source is connected to the MiniPLC terminals V+ and 0V.**

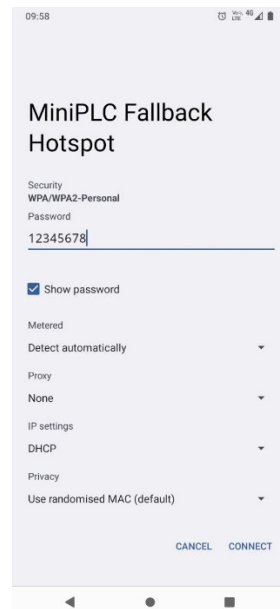
Connecting to the wireless network

To link the MiniPLC to your home Wi-Fi network, start by powering up the MiniPLC, and then use your mobile device or laptop:

1. Connect to "MiniPLC Fallback Hotspot"

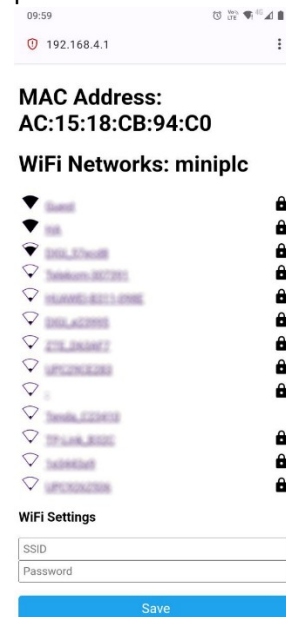


2. Log in using the password "12345678"

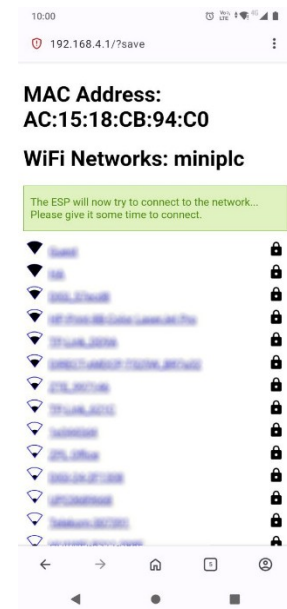


3. Access the URL <http://192.168.4.1>

Choose your home Wi-Fi network and input the password



4. After pressing Save, you can view the assigned address of the MiniPLC in your router's interface.



Web Server

With the factory firmware settings, the MiniPLC includes a built-in web server. You can access it through your web browser by entering the address that your Wi-Fi router has assigned to the MiniPLC.

Build in webserver:

MiniPLC

started 29 minutes 35 seconds ago

Sensor and Control

1-WIRE Dallas temperature BUS1

29.1 °C

1-WIRE Dallas temperature BUS2

29.5 °C

ADC AI1

8.410 V

ADC AI2

0.056 V

ADC AI3

0.046 V

ADC AI4

0.046 V

Button #1

Button #2

Button #3

Button #4

Current time

2025-04-07 15:09

DAC 0-10V

100 0 100

DI #1

DI #2

DI #3

DI #4

LED #2

LED #3

MAX 31856 Temperature 1

22.41 °C

MAX 31856 Temperature 2

23.30 °C

Modbus Sensor

11 C

RELAY #1

RELAY #2

RELAY #3

RELAY #4

RELAY #5

RELAY #6

Switch buzzer

OTA Update

Choose File

NO FILE CHOSEN

UPDATE

Debug Log

Time	Level	Tag	Message
18:09:18	[D]	[modbus_controller.sensor:025]	Sensor new state: 11.00
18:09:18	[D]	[sensor:094]	'Modbus Sensor': Sending state 11.00000 C with 0 decimals of accuracy
18:09:18	[D]	[ads1115.sensor:017]	'ADC AI2': Got Voltage=0.018175V
18:09:18	[D]	[sensor:094]	'ADC AI2': Sending state 0.05512 V with 3 decimals of accuracy
18:09:18	[D]	[ads1115.sensor:017]	'ADC AI3': Got Voltage=0.015000V
18:09:18	[D]	[sensor:094]	'ADC AI3': Sending state 0.04500 V with 3 decimals of accuracy
18:09:19	[D]	[ads1115.sensor:017]	'ADC AI1': Got Voltage=2.004625V
18:09:19	[D]	[sensor:094]	'ADC AI1': Sending state 8.41388 V with 3 decimals of accuracy
18:09:19	[D]	[modbus_controller.sensor:025]	Sensor new state: 11.00
18:09:19	[D]	[sensor:094]	'Modbus Sensor': Sending state 11.00000 C with 0 decimals of accuracy
18:09:19	[D]	[ads1115.sensor:017]	'ADC AI4': Got Voltage=0.015000V
18:09:19	[D]	[sensor:094]	'ADC AI4': Sending state 0.04500 V with 3 decimals of accuracy
18:09:19	[D]	[dallas.temp.sensor:054]	'1-WIRE Dallas temperature BUS2': Got temperature=29.7°C
18:09:19	[D]	[sensor:094]	'1-WIRE Dallas temperature BUS2': Sending state 29.68750 °C with 1 decimals of accuracy
18:09:19	[D]	[dallas.temp.sensor:054]	'1-WIRE Dallas temperature BUS1': Got temperature=29.1°C
18:09:19	[D]	[sensor:094]	'1-WIRE Dallas temperature BUS1': Sending state 29.12500 °C with 1 decimals of accuracy
18:09:20	[D]	[modbus_controller.sensor:025]	Sensor new state: 11.00
18:09:20	[D]	[sensor:094]	'Modbus Sensor': Sending state 11.00000 C with 0 decimals of accuracy
18:09:21	[D]	[modbus_controller.sensor:025]	Sensor new state: 11.00
18:09:21	[D]	[sensor:094]	'Modbus Sensor': Sending state 11.00000 C with 0 decimals of accuracy
18:09:22	[D]	[modbus_controller.sensor:025]	Sensor new state: 11.00
18:09:22	[D]	[sensor:094]	'Modbus Sensor': Sending state 11.00000 C with 0 decimals of accuracy
18:09:22	[D]	[text_sensor:064]	'Current time': Sending state '2025-04-07 15:09'
18:09:23	[D]	[modbus_controller.sensor:025]	Sensor new state: 11.00
18:09:23	[D]	[sensor:094]	'Modbus Sensor': Sending state 11.00000 C with 0 decimals of accuracy
18:09:23	[D]	[ads1115.sensor:017]	'ADC AI2': Got Voltage=0.018562V
18:09:23	[D]	[sensor:094]	'ADC AI2': Sending state 0.05569 V with 3 decimals of accuracy
18:09:23	[D]	[ads1115.sensor:017]	'ADC AI3': Got Voltage=0.015188V
18:09:23	[D]	[sensor:094]	'ADC AI3': Sending state 0.04556 V with 3 decimals of accuracy
18:09:24	[D]	[ads1115.sensor:017]	'ADC AI1': Got Voltage=2.003313V
18:09:24	[D]	[sensor:094]	'ADC AI1': Sending state 8.40994 V with 3 decimals of accuracy
18:09:24	[D]	[modbus_controller.sensor:025]	Sensor new state: 11.00
18:09:24	[D]	[sensor:094]	'Modbus Sensor': Sending state 11.00000 C with 0 decimals of accuracy
18:09:24	[D]	[ads1115.sensor:017]	'ADC AI4': Got Voltage=0.015188V
18:09:24	[D]	[sensor:094]	'ADC AI4': Sending state 0.04556 V with 3 decimals of accuracy
18:09:24	[D]	[dallas.temp.sensor:054]	'1-WIRE Dallas temperature BUS2': Got temperature=29.5°C
18:09:24	[D]	[sensor:094]	'1-WIRE Dallas temperature BUS2': Sending state 29.50000 °C with 1 decimals of accuracy
18:09:24	[D]	[dallas.temp.sensor:054]	'1-WIRE Dallas temperature BUS1': Got temperature=29.1°C
18:09:24	[D]	[sensor:094]	'1-WIRE Dallas temperature BUS1': Sending state 29.06250 °C with 1 decimals of accuracy
18:09:25	[D]	[modbus_controller.sensor:025]	Sensor new state: 11.00
18:09:25	[D]	[sensor:094]	'Modbus Sensor': Sending state 11.00000 C with 0 decimals of accuracy

Connecting to the Home Assistant platform

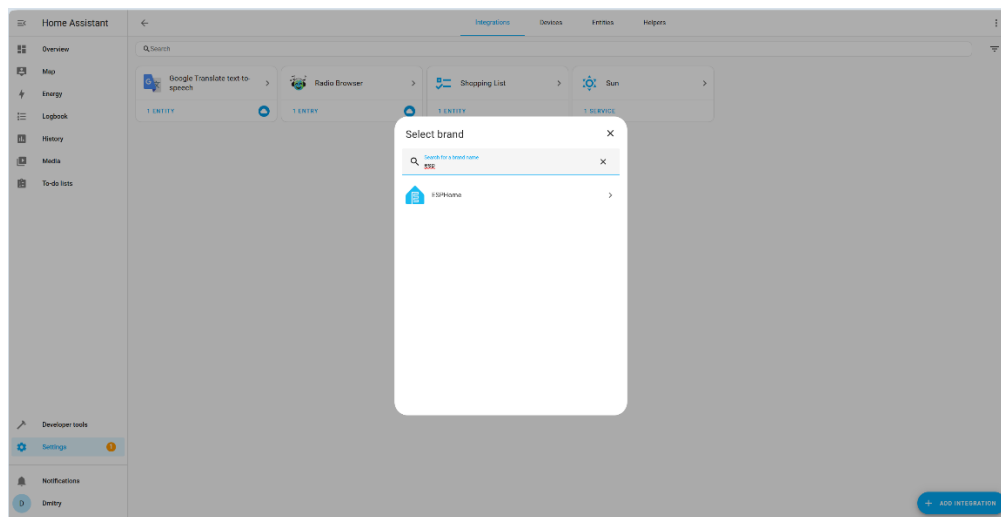
To connect the MiniPLC to your Home Assistant setup, follow these steps:

1.Power the MiniPLC

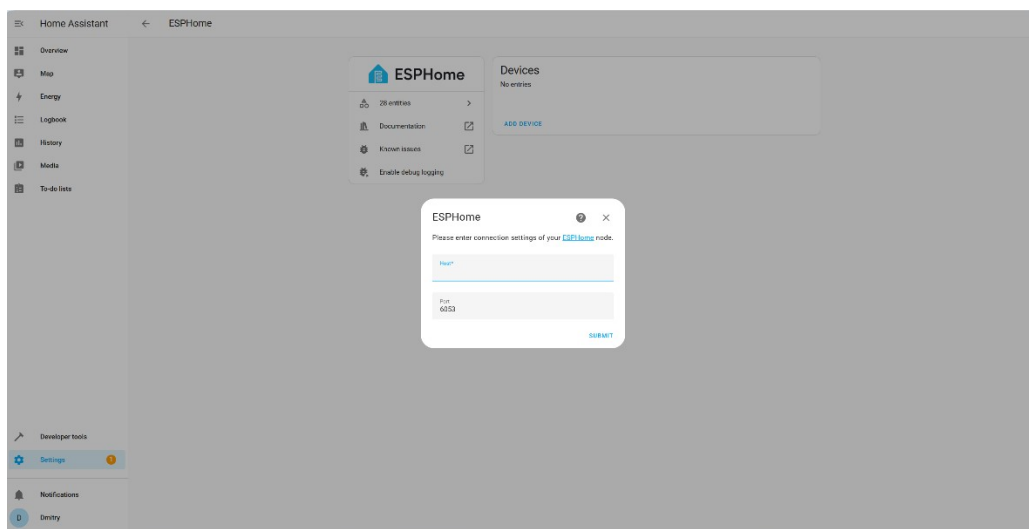
2.Connect to Wi-Fi

3.Integrate with Home Assistant:

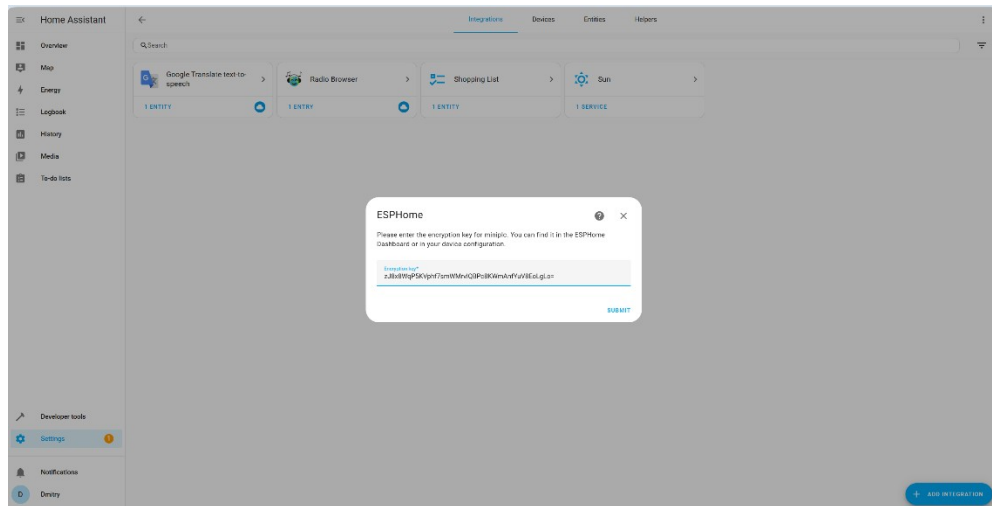
Setup ESPHome integration in Home Assistant:



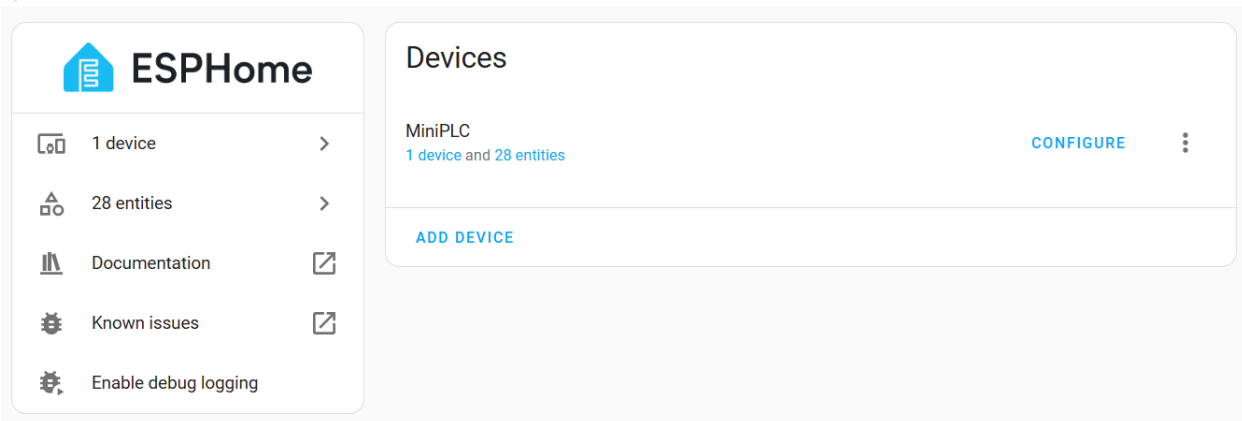
In the ESPHome integration press "Add device" and enter the address that your Wi-Fi router has assigned to the MiniPLC:



Input the encryption key from your configuration file, or the default encryption key provided in the factory setup firmware "zJ8x8WqP5KVphf7smWMrvlQBpc8KWmAnfYuV8EoLgLo=":



By following these steps, you can effectively connect and integrate the ESP32 MiniPLC with your Home Assistant system:



Controls

	DAC 0-10V	☐
	LED #2	☐
	LED #3	☐
	RELAY #1	☐
	RELAY #2	☐
	RELAY #3	☐
	RELAY #4	☐
	RELAY #5	☐
	RELAY #6	☐
	Switch buzzer	☒

ADD TO DASHBOARD

Sensors

	1-WIRE Dallas temperature...	29.1 °C
	1-WIRE Dallas temperature...	29.5 °C
	ADC AI1	8.412 V
	ADC AI2	0.055 V
	ADC AI3	0.044 V
	ADC AI4	0.047 V
	Button #1	Off
	Button #2	Off
	Button #3	Off
	Button #4	Off
	Current time	2025-04-07 15:01
	DI #1	Off
	DI #2	Off
	DI #3	Off
	DI #4	Off
	MAX 31856 Temperature 1	22.45 °C
	MAX 31856 Temperature 2	23.36 °C
	Modbus Sensor	11 C

ADD TO DASHBOARD

Firmware Download

Pre-installed Firmware:

The ESP32 MiniPLC comes with ESPHome firmware pre-installed, meaning it's ready for integration with Home Assistant right out of the box.

Reprogramming MiniPLC: You can reprogram or update the settings via ESPHome using the USB Type-C interface, wireless or manually via OTA in the embedded web server.

Downloading firmware via USB:

In the ESPHome dashboard add a New device

Skip the test:

- "MiniPLC"

New device

Name*

MiniPLC

This device will be configured to connect to the Wi-Fi network stored in your secrets.

CANCELNEXT

Installation

ESPHome will now create your configuration and install it on your device.

Connect your ESP8266 or ESP32 with a USB cable to your computer and click on connect. You need to do this once. Later updates install wirelessly. [Learn more](#)

Skip this step to install it on your device later or if you are using a Raspberry Pi Pico.

SKIP THIS STEPCONNECT

Select the device type - ESP32:

Select your device type

Select the type of device that this configuration will be installed on.

ESP32

>

ESP32-S2

>

ESP32-S3

>

ESP32-C3

>

ESP8266

>

Raspberry Pi Pico W

>

BK72xx

>

RTL87xx

>

☒ Use recommended settings

CANCEL

Choose skip installation:



Configuration created!

You can now install the configuration to your device. The first time this requires a cable.

Once the device is installed and connected to your network, you will be able to manage it wirelessly.

Each ESPHome device has a unique encryption key to talk to other devices. You will need this key to include your device in Home Assistant. You can find the key later in the device menu.

Encryption key

/8soHonSOBc4FiFRcjRivZXHlsRyGwYzIK2g=

SKIPINSTALL

<https://github.com/isystemsautomation/HOMEMASTER/tree/main/Firmware/MiniPLC>):

```

174 x_minipyclam
175   reference_resistance: 9000 Q
176   rtd_nominal_resistance: 1000 Q
177   update_interval: 60s
178   platform: adxl115
179   multiplatform: "AD_ZERO"
180   gain: 6.144
181   name: "ADC_AZ0"
182   update_interval: 5s
183   filters:
184     multiply: 1
185     platform: adxl115
186     multiplatform: "AD_ZERO"
187     gain: 6.144
188     name: "ADC_AZ1"
189     update_interval: 5s
190     filters:
191       multiply: 1
192     platform: adxl115
193     multiplatform: "AD_ZERO"
194     gain: 6.144
195     name: "ADC_AZ2"
196     update_interval: 5s
197     filters:
198       multiply: 1
199     platform: adxl115
200     multiplatform: "AD_ZERO"
201     gain: 6.144
202     name: "ADC_AZ3"
203     update_interval: 5s
204     filters:
205       multiply: 1
206     platform: mcp3202
207     multiplatform: "AD_ZERO"
208     name: "Modbus Sensor"
209     register_type: holding
210     address: 0x0001 # Address of the register inside the Modbus slave device
211     unit_of_measurement: "C"
212     value_type: U_WORD
213
214 output:
215   platform: i2c
216   pin: GP02
217   i2c_buzzer_output
218   platform: mcp47xx
219   id: dac_output
220   status_led:
221     pin:
222       pc185M1: pin185M1 hub.a
223     number: 7
224     mode:
225       output: true
226     inverted: false

```

Install the settings through a USB cable:

The screenshot shows the ESPHome Web interface. On the left, a sidebar lists four connection methods: 'Wirelessly' (requires device to be online), 'Plug into this computer' (highlighted in grey, for devices connected via USB), 'Plug into the computer running ESPHome Device Builder' (for devices connected via USB to the server), and 'Manual download' (install it yourself using ESPHome Web or other tools). A 'CANCEL' button is at the bottom left. The main area displays the URL 'http://127.0.0.1:6052 wants to connect to a serial port' and a list of available ports, with 'CP2102N USB to UART Bridge Controller (COM5) - Paired' selected. At the bottom right are 'Connect' and 'Cancel' buttons. An inset window on the right shows a progress bar at 4% and the text 'Installing. This will take 2 minutes. Keep this page visible to prevent slow down'.

Wiring diagrams:

1-WIRE

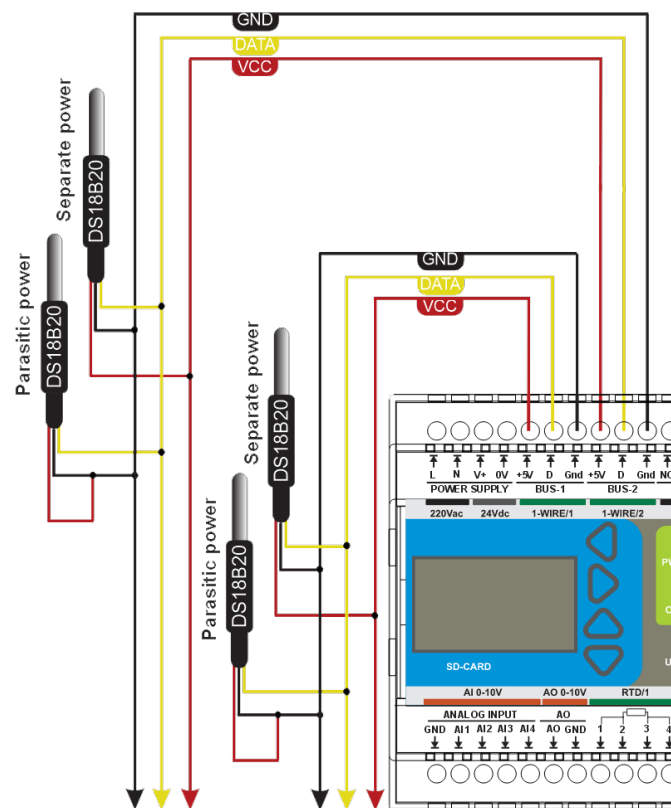
It is advisable to connect a maximum of 20 sensors to the 1-wire bus for optimal performance.

While it is technically feasible to connect up to 32 sensors, doing so will necessitate more complex wiring, which may result in reduced cable length and longer polling intervals.

For the signal lines DATA and GND, utilize a twisted pair from a shielded CAT5/6/7 cable. The VDD line can be connected to an additional conductor within the same cable.

Please ensure that the cable shield remains unconnected, and refrain from transmitting any other data signals through the same cable.

Connection Diagram:



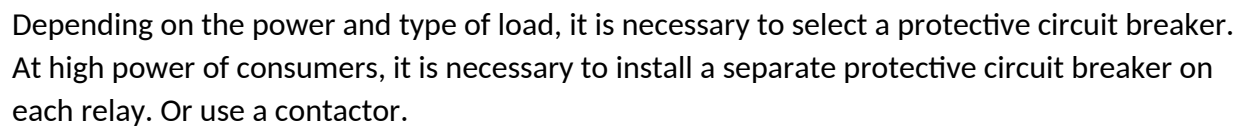
The maximum permissible length of a 1-wire bus is contingent upon the chosen wiring topology. In a linear configuration and a bus topology with short taps, the length can extend up to 300 meters. Conversely, for bus topologies featuring long taps and star configurations, the

Numerous 1-wire sensors are designed to operate with parasitic power supply, allowing them to draw power through the data line (DATA) while the sensor's VDD is linked to GND. This configuration enables the elimination of one conductor, albeit at the cost of the maximum allowable length of the 1-wire bus.

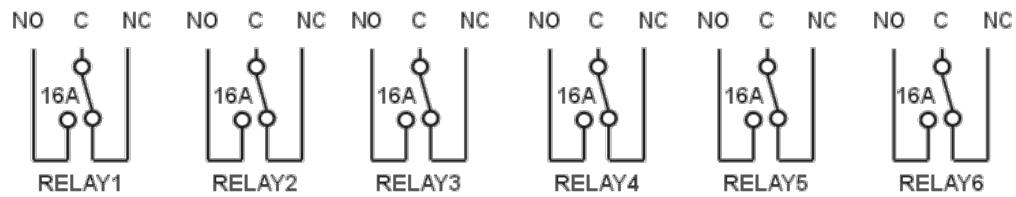
MiniPLC equipped with 6 digital outputs (incorporating relays with optocoupler and varistor for enhanced surge protection)

dry contact specifications:

- Connection example:



Relay contacts inside MiniPLC:



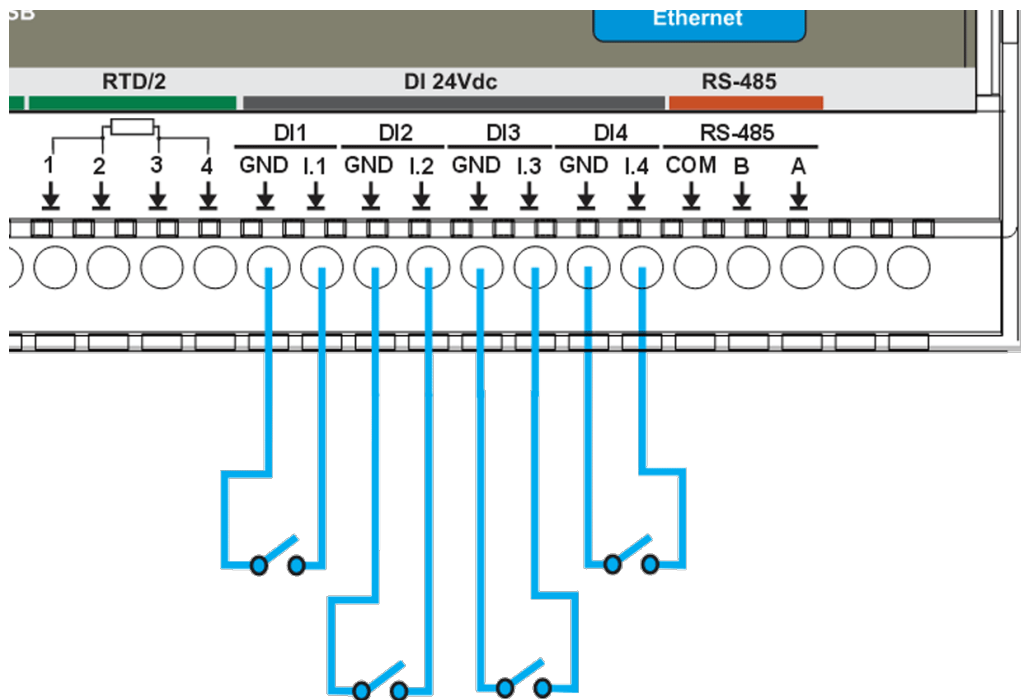
C* Each relay contact is protected from overvoltage by a varistor connected to contact "C"

Digital Inputs

MiniPLC equipped with 4 isolated, surge protected digital sourcing inputs:

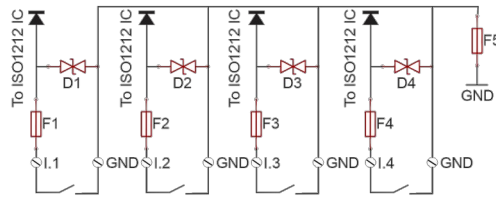
- logic 0: 0 ... 9.2VDC
- undefined: 9.2 ... 15.8VDC
- logic 1: 15.8 ... 24VDC

Connection example:



Digital inputs are designated as source inputs. It is important to note that digital input channels are not intended to serve as a power supply for any devices

Fuses and TVS diodes within the MicroPLC:

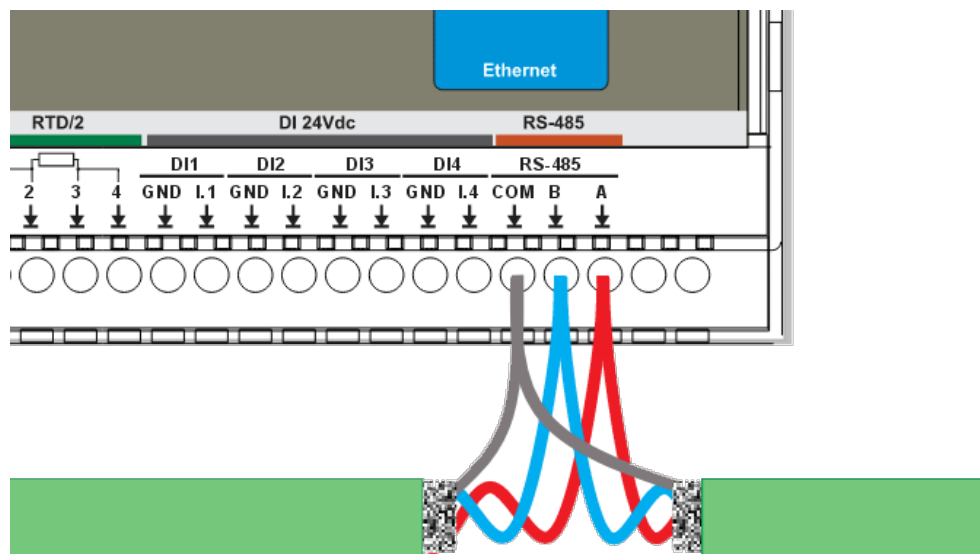


RS-485

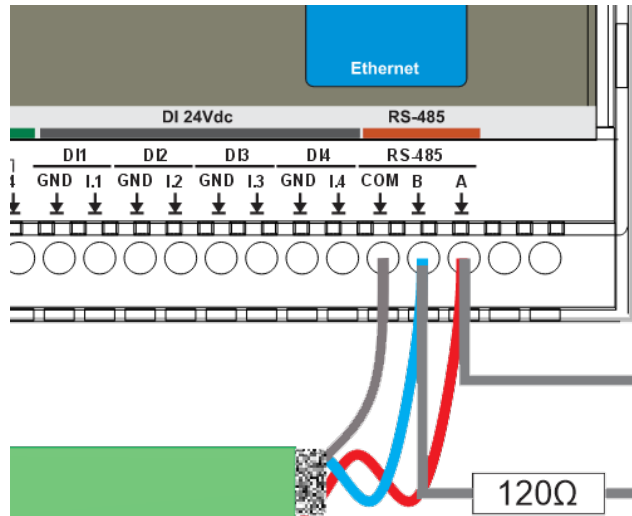
The RS-485 port is not galvanically isolated (MAX485; transceiver shares logic ground). Run COM to every node.

In RS-485 communication systems, it is essential to utilize a shielded twisted pair cable to ensure optimal performance. A notable example of such a cable is the Belden 3105A, compliant with standards EN:50290-2-23 and IEC:60502. However, any cable exhibiting comparable specifications can effectively facilitate the interconnection of all devices within the network.

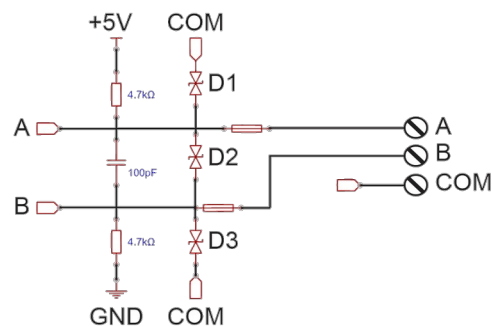
Connection example:



To avoid signal reflections, it is necessary to connect 120 Ohm termination resistors to the ends of the communication cable.:



Automatic fuses and TVS diodes inside MiniPLC for protection:

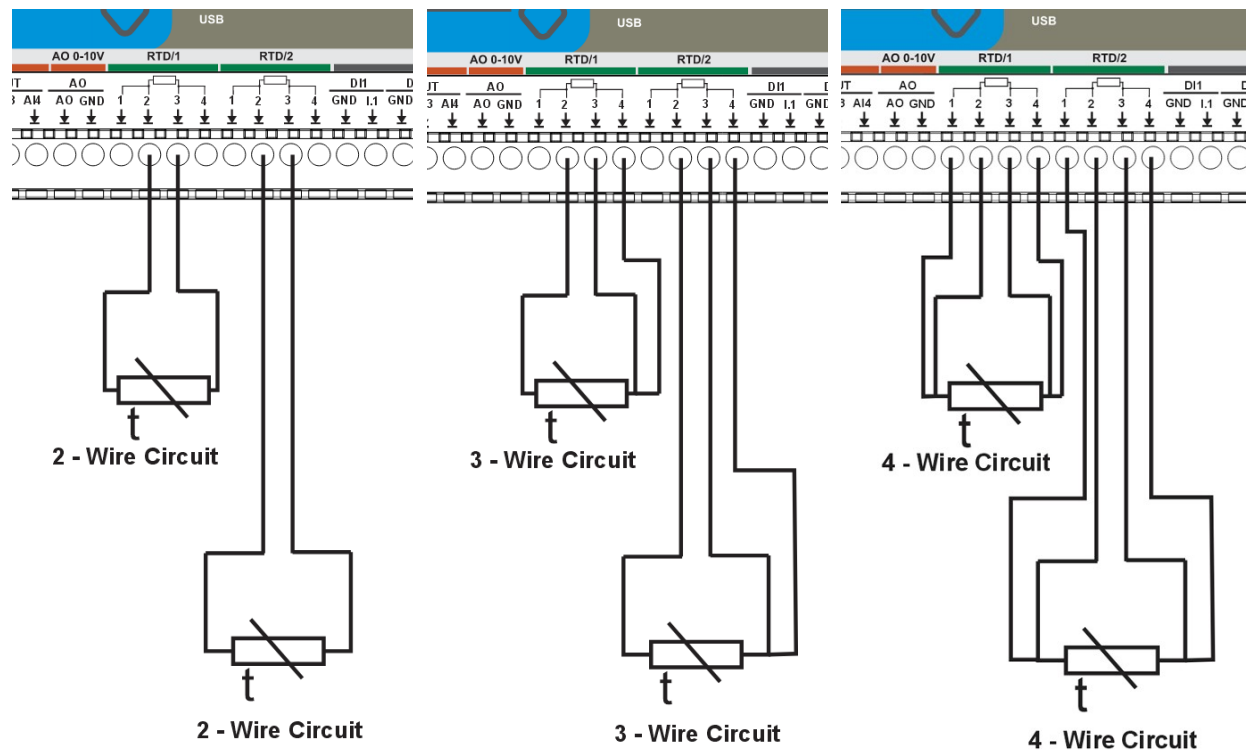


RTD inputs

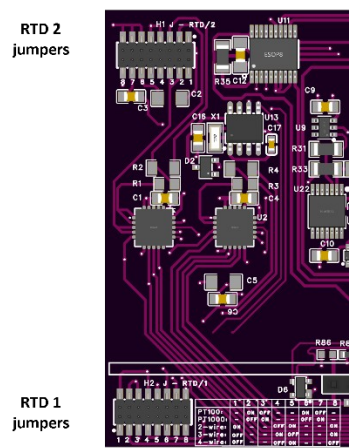
MiniPLC equipped with the MAX31865, capable of measuring 100Ω to 1kΩ Platinum RTDs (PT100 to PT1000) at 0°C. It supports 2-, 3-, and 4-wire sensor connections, features a 15-bit ADC resolution, and offers a nominal temperature resolution of 0.03125°C (which may vary due to RTD nonlinearity). The total accuracy under all operating conditions is a maximum of 0.5°C (0.05% of full scale). It includes fully differential VREF inputs, a maximum conversion time of 21ms, and ±45V input protection.

Additionally, it provides fault detection for open RTD elements, shorted RTDs, out-of-range voltage, or shorts across the RTD element

Connection example:



To configure the RTD for 2, 3, or 4 wire circuits with either PT100 or PT1000 types, it is essential to adjust the jumper settings on the MCU board:



The factory's default jumper setup is configured for a 2-wire PT100 RTD type for the RTD/1 channel, and a 2-wire PT1000 RTD type for the RTD/2 channel:

	J1	J2	J3	J4	J5	J6	J7	J8
--	----	----	----	----	----	----	----	----

PT100	-	ON	OFF	-	-	ON	OFF	-
PT1000	-	OFF	ON	-	-	OFF	ON	-
2-WIRE	ON	-	-	OFF	ON	-	-	ON
3-WIRE	OFF	-	-	ON	OFF	-	-	ON
4-WIRE	OFF	-	-	OFF	ON	-	-	OFF

Additionally, it is essential to configure the reference resistor in the YAML file:

#MAX 31865 RTD

- platform: max31865

name: "MAX 31856 Temperature 1"

cs_pin: GPIO1

#Reference resistance for PT100

reference_resistance: 400 Ω

rtd_nominal_resistance: 100 Ω

update_interval: 60s

- platform: max31865

name: "MAX 31856 Temperature 2"

cs_pin: GPIO3

#Reference resistance for PT1000

reference_resistance: 4000 Ω

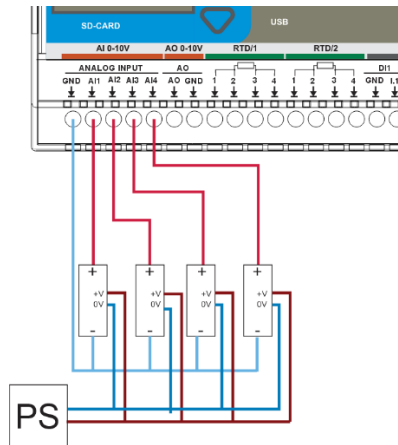
rtd_nominal_resistance: 1000 Ω

update_interval: 60s

0-10V Analog inputs

MiniPLC integrated with the ADS1115, providing 4 analog inputs from 0 to 10VDC, featuring a 16-bit resolution through an operational amplifier, a maximum sampling rate of 0.86 ksp/s, ESD ratings of 2kV HBM and 1.5kV CDM, and includes EMI and RF filtering.

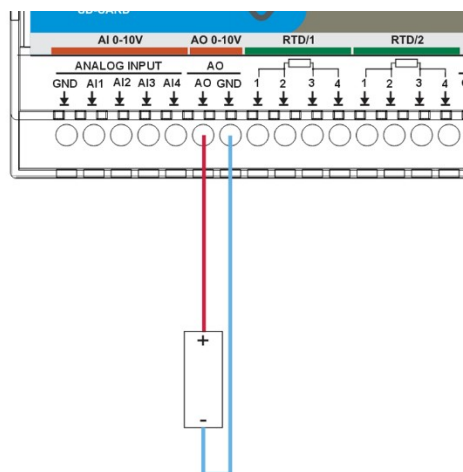
Connection example:



0-10V Analog output

A MiniPLC combined with the MCP4725, delivering an analog output range of 0 to 10VDC, with a resolution of 12 bits achieved through an operational amplifier.

Connection example:



Bus system configuration

I2C:

- sda: GPIO32
- scl: GPIO33

I2C addresses:

- pcf8574/2: 0x38
- pcf8574/1: 0x39
- ads1115: 0x48
- SH1106 128x64: 0x3C
- pcf8563: 0x51

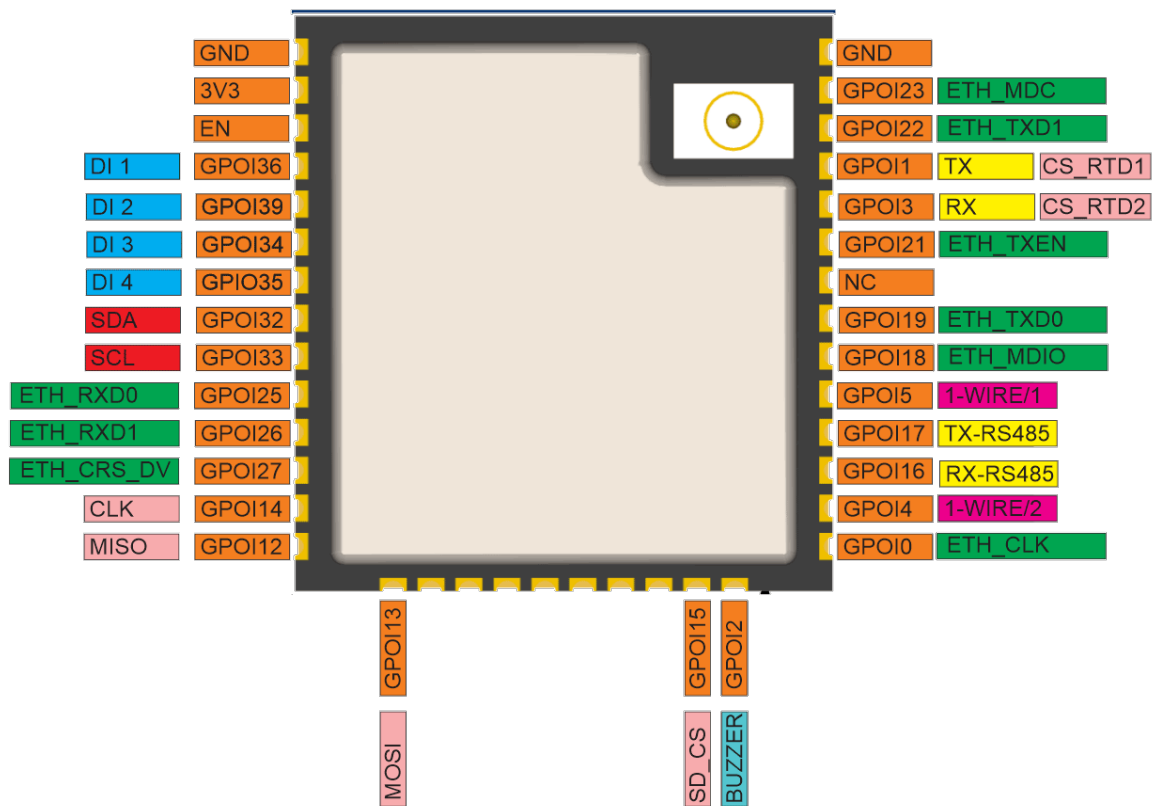
SPI:

- miso_pin: GPIO12
- mosi_pin: GPIO13
- clk_pin: GPIO14

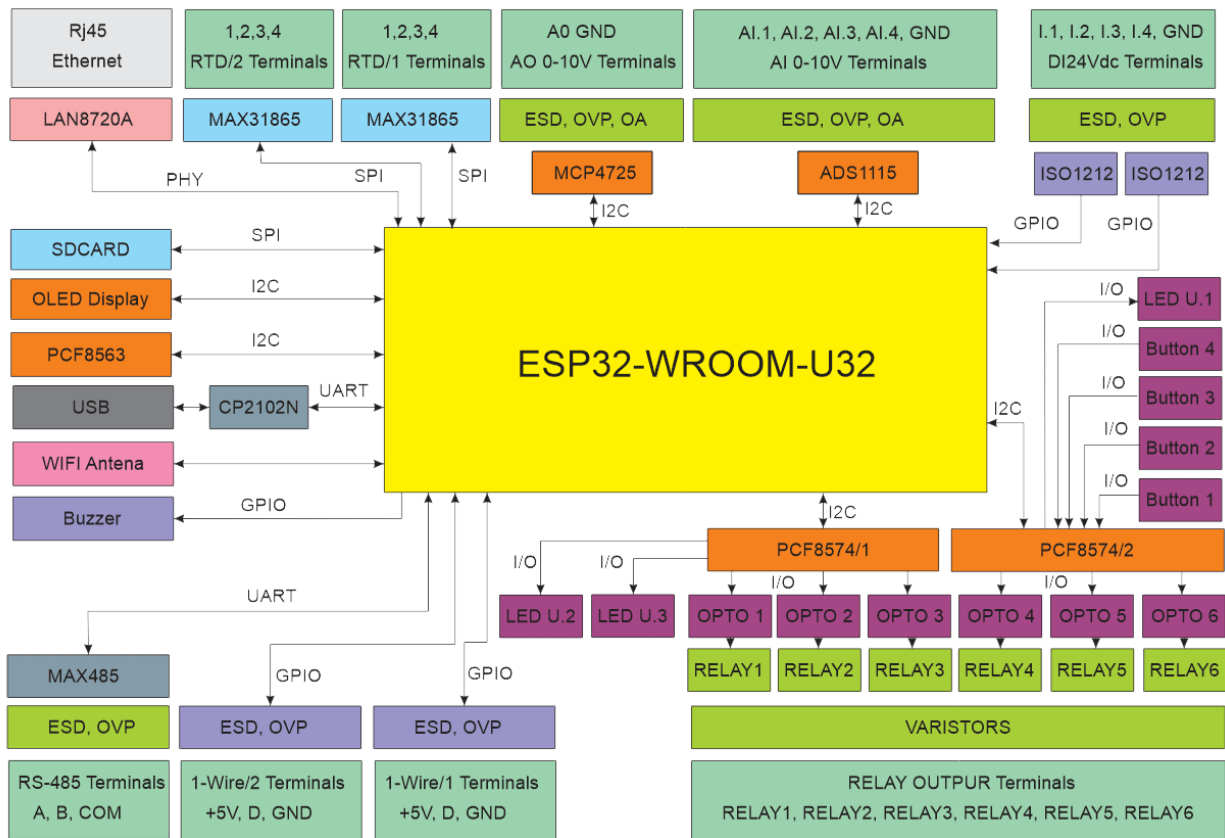
SPI CS pins:

- max31865 RTD1: GPIO01
- max31865 RTD2: GPIO03
- SDCARD: GPIO15

ESP32 Pinout



MiniPLC Function Block Diagram



MiniPLC boards

